Linguistic MDL Explanations with LLMs

via Occam-Style Prompt Discovery

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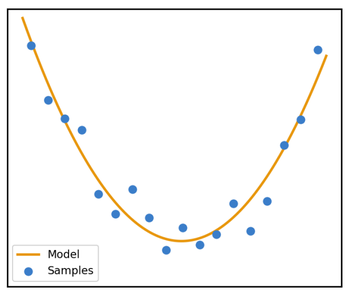
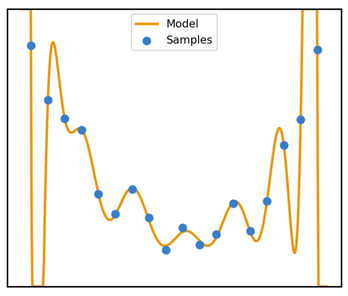
Motivation & MDL Principle





Which explanation would you trust?

Most people prefer the simple quadratic fit (left) over the highly complex curve (right), even though both match the data.

Why? We naturally apply *Occam's razor*: prefer explanations that are both simple and predictive.

But language models do not explicitly optimize for this. They favor explanations that are *likely to be said*, not necessarily those that best explain the data.

We formalize this intuition using the **Minimum Description Length (MDL)** principle, selecting an explanation z that minimizes:

$$\mathcal{L}_{\text{MDL}}(z, x) = \underbrace{-\log p^*(z)}_{\text{complexity}} + \underbrace{-\log p^*(x \mid z)}_{\text{data fit}}$$

- Simple explanations → lower complexity cost
- Predictive explanations → lower data cost
- Best explanation → shortest total description

Method

We use a pretrained LLM in two roles: a frozen evaluator p^* and a trainable policy $\pi_\theta(z \mid x)$ (via LoRA).

Given data x, the policy samples an explanation z, which is scored by an MDL-style reward:

$$R(z, x) = \lambda \log p^*(z) + \log p^*(x \mid z).$$

The first term favors concise explanations, while the second favors explanations that predict the data.

The policy is optimized with REINFORCE:

$$\mathcal{L} = -(R - b) \log \pi_\theta(z \mid x),$$

where b is a baseline.

Key idea: learn a policy that proposes explanations maximizing simplicity and predictive accuracy under a fixed evaluator.

Guess the Loss!

Rank the explanations from **lowest MDL loss** (best) to highest.

Sequence A:
$1, 1, 2, 5, 12, 27, 58, 121, \ldots$

1. $2^n - n$
2. $a_n = a_{n-1} + a_{n-2}$
3. `the pattern is [1, 1, 2, 5, 12, 27, 58, 121]`

Sequence B:
$6, 7, 13, 20, 33, 53, 86, 139, \ldots$

1. $a_n = a_{n-1} + a_{n-2}$
2. $a_n = 2n + 5$
3. `the pattern is [6, 7, 13, 20, 33, 53, 86, 139]`

Answer (MDL ranking):

- **A:** $2^n - n <$ Fibonacci $<$ memorization
- **B:** Fibonacci $<$ linear $<$ memorization

MDL prefers explanations that are both simple and predictive, and penalizes memorization.

Prompt Templates

System: You are a careful pattern-discovery assistant. Prefer concise, mathematical rules when possible.

Prior term: $-\log p(z)$

```
User: Write a concise rule for a number sequence.
Assistant: {z}
```

Likelihood term: $-\log p(x \mid z)$

```
User: Rule: {z}
Output EXACTLY the first 10 terms as comma-separated integers.
Assistant: {x}
```

Conditional baseline: $-\log p(z \mid x)$

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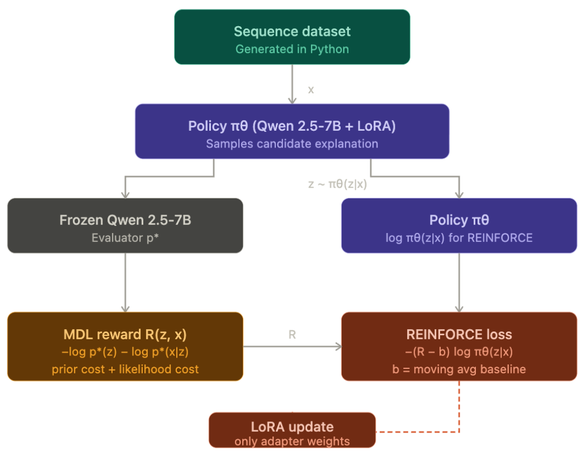
User: Observed sequence: 1, 1, 2, 5, 12, 27, 58, 121, 248, 503
What is the most likely rule?
Assistant: {z}
```

Training Loop



- We generate a synthetic dataset of number sequences x in Python.
- A LoRA-adapted policy $\pi_\theta(z \mid x)$ samples candidate explanations z.
- A frozen evaluator p^* scores each explanation using the MDL objective:

$$-\log p^*(z) - \log p^*(x \mid z)$$

- The same sampled explanation is also scored under the policy via $\log \pi_\theta(z \mid x)$.
- We combine both terms in a REINFORCE loss to update only the LoRA adapter weights.

Challenges

- **Likelihood estimation:** Accurately computing $p(x \mid z)$ is difficult with small LMs. Without chain-of-thought, models struggle with precise reasoning (e.g., arithmetic), leading to noisy likelihood estimates.
- **Scaling to natural language:** On real-world data (e.g., ScienceWorld-style traces), MDL scores become less meaningful. Token-level probabilities are diffuse, and differences outside the top predictions provide weak signal for explanation quality.
- **Exploration / discovery:** Correct explanations often lie in low-probability regions of the model. Policy optimization improves local candidates but rarely discovers the true rule without guidance.
- **Ablation (oracle injection):** Injecting the correct rule confirms that the reward and update can reinforce it once observed, indicating that the main bottleneck is exploration rather than reward design.

Results

We compare explanations using three scores: base-model conditional likelihood, fine-tuned conditional likelihood, and frozen-evaluator MDL ($\lambda = 0.3$).

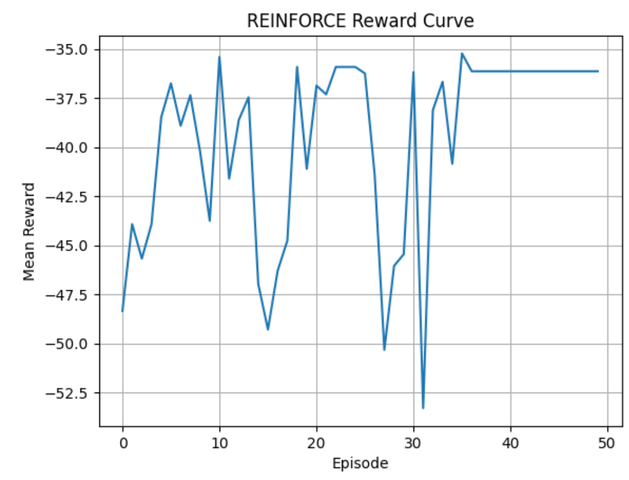
Explanation z	Base cond.	FT cond.	MDL
Base model greedy output	**2.36**	–	47.87
Fine-tuned greedy output	–	**0.13**	38.38
$2^n - n$ (true rule)	39.35	**0.09**	**29.38**
Max Likelihood	**2.36**	34.56	47.87
Dummy (copy data)	63.88	62.52	95.77

Table 1. Lower is better. MDL uses the frozen evaluator. Greedy decoding need not recover the globally lowest conditional-loss explanation.

Key observation:

- The base model prefers the max-likelihood recurrence rule.
- After fine-tuning, the policy assigns low conditional loss to the true rule.
- Fine-tuning moves model preference closer to the MDL objective.

Training: We fine-tune a LoRA policy using REINFORCE with an MDL reward. To improve exploration, we inject the correct rule with small probability during sampling.



Training exhibits high variance due to stochastic sampling, but converges to a stable reward.

Before vs After

Observed sequence:
$(1, 1, 2, 5, 12, 27, 58, 121, 248, 503)$

Before (Max Likelihood)
Explanation:
Each term is the sum of the previous term and twice the term before it.
Conditional loss: 2.36
MDL loss: 46.69

Fluent but incorrect

After (MDL fine-tuned)
Explanation:
$2^n - n$
Conditional loss: **0.23**
MDL loss: **29.38**

Concise and correct

Conclusion

MDL-based fine-tuning (via LoRA) shifts LLMs from producing likely text to generating better explanations by balancing plausibility $p(z)$ and predictive accuracy $p(x \mid z)$. This leads to concise, interpretable, and more correct rules, and extends beyond number sequences to general pattern discovery.

References

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